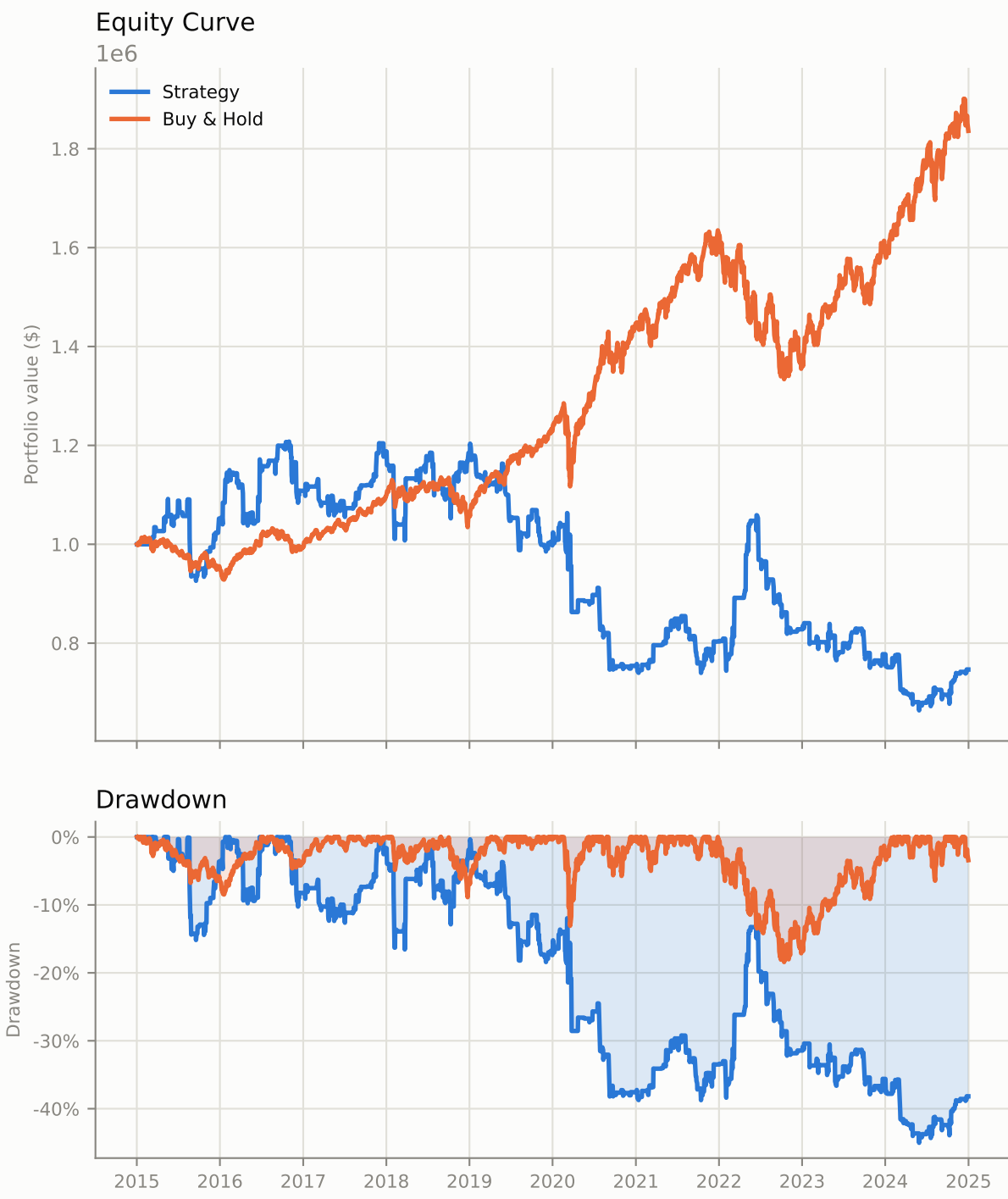


Performance Overview - Vol Sized



Metric	Strategy	Buy & Hold
Total return	-25.34%	83.62%
Annualized return	-2.88%	6.27%
Annualized vol	15.01%	8.55%
Sharpe	-0.25	0.52
Max drawdown	-45.02%	-18.41%
Drawdown/vol	3.00	2.15
Num trades	699	8
Win rate	56.80%	62.50%
Avg win	7,244.29	183,135.61
Avg loss	-10,362.10	-26,577.97
Risk/reward ratio	0.70	6.89
Payoff factor	0.92	11.48
CPC index	0.36	49.46
Time in market	27.31%	99.96%
Annual turnover	163.3x	0.2x



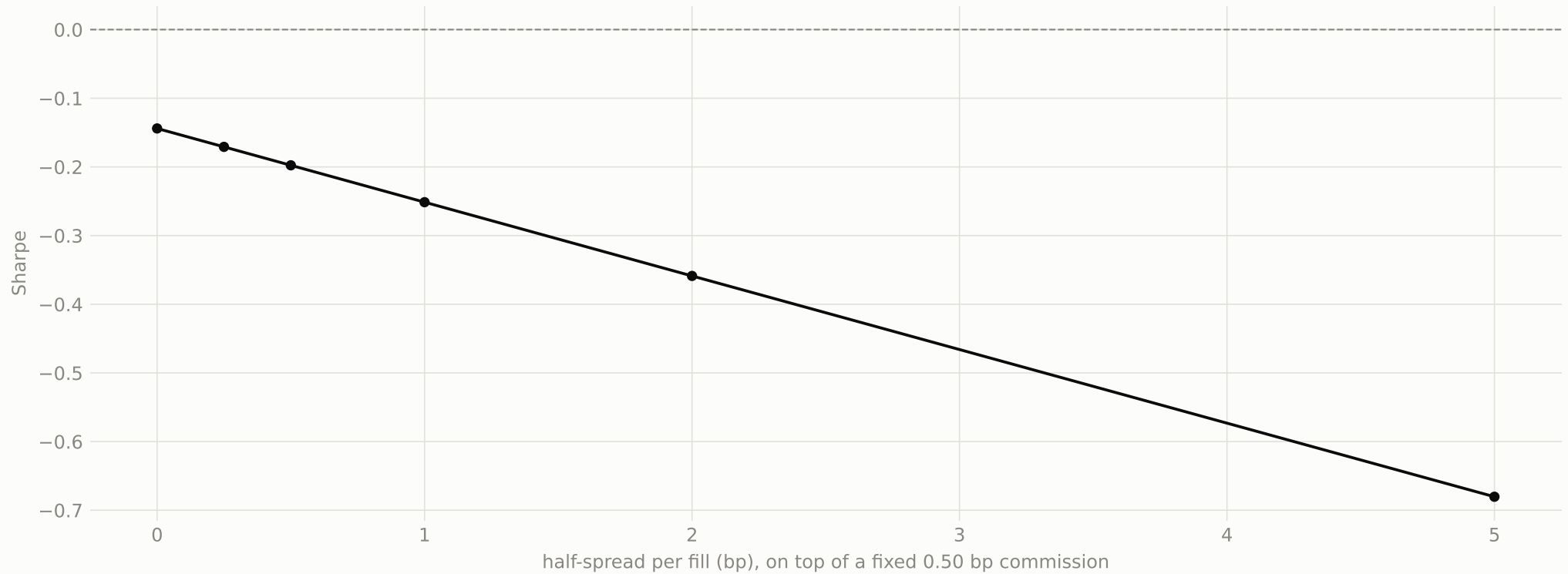
Monthly Returns

Strategy															
2015	0.0%	0.0%	2.7%	0.0%	3.2%	0.3%	-0.5%	-11.4%	1.2%	0.7%	4.2%	5.9%	5.2%	-15.2%	0.44
2016	7.5%	1.3%	-2.1%	-6.7%	-0.4%	10.3%	0.7%	1.0%	2.6%	0.7%	-3.5%	-4.9%	5.4%	-10.2%	0.41
2017	0.8%	0.0%	-2.9%	-1.4%	1.0%	-0.6%	0.0%	2.1%	2.1%	0.9%	6.7%	-1.4%	7.2%	-6.8%	0.83
2018	-2.4%	-10.3%	9.0%	-0.0%	1.3%	2.6%	-5.3%	1.2%	-1.7%	2.4%	2.5%	1.7%	-0.3%	-15.2%	0.08
2019	-3.6%	-0.5%	-1.5%	0.3%	2.9%	-9.4%	0.6%	-3.0%	3.0%	-2.7%	-2.9%	0.3%	-15.8%	-18.1%	-1.46
2020	3.6%	-4.4%	-12.7%	2.7%	-0.2%	1.4%	-7.9%	-0.8%	-8.7%	1.2%	-0.7%	0.2%	-25.1%	-29.8%	-1.19
2021	0.3%	0.1%	5.2%	0.1%	3.7%	2.8%	0.6%	-5.0%	-6.3%	2.1%	0.7%	2.8%	6.5%	-13.5%	0.58
2022	-4.1%	5.3%	9.9%	9.8%	7.0%	-7.5%	-6.0%	-3.3%	-0.3%	-5.4%	-0.9%	0.6%	3.1%	-22.7%	0.25
2023	1.5%	-4.6%	0.1%	1.0%	-5.5%	4.5%	0.2%	3.4%	-3.6%	-5.2%	1.0%	1.7%	-6.1%	-10.8%	-0.42
2024	-3.4%	3.4%	-9.5%	-0.8%	-2.8%	0.4%	2.0%	1.8%	-1.4%	4.1%	2.4%	0.6%	-4.0%	-14.6%	-0.29
													Annual	Max DD	Sharpe

Buy & Hold															
2015	1.1%	0.0%	-1.5%	0.6%	-0.2%	-1.5%	-0.8%	-1.4%	-0.9%	2.5%	-1.6%	-1.3%	-5.0%	-6.7%	-0.91
2016	-0.2%	1.5%	1.8%	0.9%	0.1%	2.4%	1.2%	-0.5%	0.5%	-1.6%	-2.0%	0.4%	4.8%	-4.6%	1.01
2017	1.6%	1.8%	-0.3%	0.8%	1.0%	-0.8%	1.6%	1.5%	-0.6%	1.4%	1.0%	0.9%	10.1%	-2.0%	2.82
2018	2.5%	-1.8%	-0.5%	0.0%	1.7%	-0.4%	0.3%	1.7%	-0.2%	-3.5%	-0.5%	-2.3%	-3.7%	-8.9%	-0.48
2019	3.8%	1.0%	1.6%	1.4%	-2.0%	3.9%	0.6%	1.5%	-0.5%	1.5%	0.8%	1.2%	15.7%	-2.3%	2.90
2020	1.7%	-2.0%	-3.5%	5.5%	2.7%	2.2%	5.0%	3.1%	-2.8%	-2.0%	4.4%	2.8%	16.8%	-13.1%	1.28
2021	-0.8%	-0.5%	0.1%	3.9%	1.0%	1.8%	2.1%	1.5%	-2.9%	4.2%	-0.1%	1.7%	12.8%	-5.0%	1.48
2022	-3.6%	-0.6%	2.1%	-5.9%	-0.4%	-5.0%	4.3%	-3.1%	-6.4%	2.1%	4.1%	-3.9%	-15.8%	-17.9%	-1.16
2023	5.3%	-2.3%	4.6%	0.6%	1.0%	2.7%	2.6%	-1.1%	-3.5%	-0.6%	5.5%	1.7%	17.5%	-5.8%	2.01
2024	0.8%	2.3%	2.5%	-2.1%	3.3%	2.8%	0.4%	1.1%	2.1%	-0.1%	2.3%	-1.7%	15.3%	-6.4%	1.59
													Annual	Max DD	Sharpe

Cost Sensitivity

Sharpe vs half-spread per fill. Commission held fixed at 0.50 bp per fill on every rung, so the whole curve already pays it.



Half-spread (bp)	Total return	Sharpe	Max drawdown
0.00	-12.3%	-0.14	-39.8%
0.25	-15.7%	-0.17	-41.1%
0.50	-19.1%	-0.20	-42.4%
1.00	-25.3%	-0.25	-45.0%
2.00	-36.5%	-0.36	-51.5%
5.00	-60.9%	-0.68	-66.9%

Annual turnover holds at 163x across the ladder: this strategy does not trade less when trading costs more, which is why a fraction of a basis point is decisive.

Backtest Configuration

Field	Value
name	Vol Sized
data	data/raw.parquet
tickers	['SPY', 'QQQ', 'TLT', 'IEF', 'GLD', 'DBC', 'UUP', 'FXE']
portfolio	inverse_vol
window	20
initial_cash	1,000,000.0
entry_threshold	2.0
exit_threshold	0.25
vol_window	20
max_gross	2.0
dollar_neutral	False
winsor_limit	3.0
cost_bps	1.0
commission_bps	0.5
drift_band	0.005
stop_loss_pct	0.04
take_profit_pct	0.15
max_holding_days	60
output_dir	output
risk_free_rate	0.02